

STAT 7500 – Statistical Programming

SAS Homework 03

Submission instructions: same as Week 02. Multiple formats are okay, so long as it is organized / well labeled.

For this homework, we will return to the *Patients Time 1* and *Patients Gender Data* context from HW 02. Recall that data were gathered on twenty patients at a clinic. The patients were given identification numbers, 001 – 020. A health awareness survey was given to the patients at baseline, which is scored out of a possible 20 points, with higher numbers indicating more awareness of health habits.

It turns out, after a six-week health awareness class, patients then took a similar health awareness survey. These data are found in the file *Patients Time 2*.

- a) Read in the *Patients Time 1*, *Patients Time 2*, and *Gender Data* datasets. Use your code from HW 02 to apply proper formatting to the gender variable. As good practice, check PROC CONTENTS and PROC PRINT to make sure everything was read in and formatted as intended.
- b) Combine the *Patients Time 1* and *Patients Time 2* into one dataset
- c) Combine the *Gender Data* into the dataset you created in part b
- d) Read in the file *Additional Patients* and combine it with the dataset from part c. Note, this file is in Excel format.
- e) Create a new variable, called *HealthAwareIncr* that is the increase on the health awareness survey score of these patients after taking the six-week course.
- f) Do a PROC FREQ of each variable and comment on what you notice.
- g) Conduct a paired t-test on the change in score by using the following code, where *PRE* is the variable you assigned to health awareness score at baseline and *POST* is the variable you assigned as their health awareness score after the six-week class. Conduct a formal hypothesis test based on this PROC, including stating the hypotheses, identifying the p-value, and a conclusion. You should also comment on whether you think the paired t-test is appropriate for these data, citing evidence when possible.
*proc ttest; paired pre*post;*
- h) Verify that the one-sample t-test of the *HealthAwareIncr* variable is the same result as the paired t-test from the previous part. What output is different?
proc ttest; var HealthAwareIncr;
- i) Compare the average increase of men vs. women using a two-sample t-test on the change in scores of men vs. women. The code is given below. Comment on what you discover.
proc ttest; var HealthAwareIncr; class gender;